

## 第八周作业参考解答

p.124, 习题 13.3, 1, 2, 3, 5, 6(1), (2), (3), (5), 7, (2), (3), 8(2), (3)

1. (1) 积分区域  $\Omega$  是一个以  $H$  为半径, 高为  $H$  的倒立圆锥, 根据三重积分的几何意义,

立体  $\Omega$  的体积为  $\iiint_{\Omega} dx dy dz = \frac{1}{3} \pi H^2 \cdot H = \frac{1}{3} \pi H^3$ .

$$(2) \iiint_{\Omega} dx dy dz = \int_0^1 \int_0^{1-x} \int_0^{1-x-y} dz dy dx = \frac{1}{6}.$$

$$2. (1) \text{ 先一后二: } \int_{-1}^1 dx \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} dy \int_{\sqrt{x^2+y^2}}^1 f(x, y, z) dz \text{ 或 } \int_{-1}^1 dy \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} dx \int_{\sqrt{x^2+y^2}}^1 f(x, y, z) dz$$

(2) 先对  $z$  积分, 再对  $y$  积分, 后对  $x$  积分:

$$I = \int_0^1 dx \int_0^{1-x} dy \int_0^{x^2+y^2} f(x, y, z) dz.$$

先对  $z$  积分, 再对  $x$  积分, 后对  $y$  积分:

$$I = \int_0^1 dy \int_0^{1-y} dx \int_0^{x^2+y^2} f(x, y, z) dz.$$

3. 解: 积分区域可以表示为:  $\Omega = \{(x, y, z) \mid 0 \leq z \leq y \leq x \leq 1\}$ ,

交换积分顺序:

$$I = \int_0^1 \frac{\cos z}{(1-z)^2} dz \int_z^1 dy \int_y^1 dx = \int_0^1 \frac{\cos z}{(1-z)^2} dz \int_z^1 (1-y) dy = \int_0^1 \frac{\cos z}{(1-z)^2} \cdot \frac{(1-z)^2}{2} dz = \frac{\sin 1}{2}.$$

5. 解: 设  $\Omega = \{(x, y, z) \mid x^2 + y^2 + z^2 \leq r^2\}$ , 在  $\Omega$  上利用三重积分的中值定理, 存在  $(\xi, \eta, \zeta) \in \Omega$  使得

$$\iiint_{x^2+y^2+z^2 \leq r^2} f(x, y, z) dx dy dz = \frac{4\pi}{3} r^3 f(\xi, \eta, \zeta).$$

由于函数  $f$  为连续函数, 因此原极限为:

$$\lim_{r \rightarrow 0^+} \frac{1}{r^3} \cdot \frac{4\pi}{3} r^3 f(\xi, \eta, \zeta) = \frac{4\pi}{3} f(0, 0, 0).$$

6. (1) 解: 由于积分区域  $\Omega$  关于  $xOz$  坐标平面对称, 因此  $\iiint_{\Omega} y dx dy dz = 0$  且

$$\iiint_{\Omega} (x+y) dx dy dz = \iiint_{\Omega} x dx dy dz.$$

将三重积分转化为先二后一的累次积分, 任取  $x \in [0, 1]$ , 记  $D_{yz} = \{(y, z) \mid \frac{y^2}{a^2} + \frac{z^2}{b^2} \leq x^2 + 1\}$ ,

则

$$\begin{aligned} \iiint_{\Omega} (x+y) dx dy dz &= \iiint_{\Omega} x dx dy dz = \int_0^1 x dx \iint_{D_{yz}} dy dz \\ &= \pi ab \int_0^1 x (\sqrt{x^2+1})^2 dx = \pi ab \int_0^1 (x^3 + x) dx = \frac{3}{4} \pi ab. \end{aligned}$$

(2) 解: 作球坐标变换:

$$\begin{cases} x = \rho \sin \varphi \cos \theta \\ y = \rho \sin \varphi \sin \theta \\ z = \rho \cos \varphi \end{cases}$$

则直角坐标系下的空间区域 $\Omega$ 变为球坐标系下的区域:

$$\Omega_{\rho\varphi\theta} = \{(\rho, \varphi, \theta) \mid 0 \leq \rho \leq 1, 0 \leq \varphi \leq \frac{\pi}{2}, 0 \leq \theta \leq \frac{\pi}{2}\},$$

故

$$\begin{aligned} \iiint_{\Omega} xyz dx dy dz &= \iiint_{\Omega_{\rho\varphi\theta}} \rho^5 \sin^3 \varphi \cos \varphi \sin \theta \cos \theta d\rho d\varphi d\theta \\ &= \int_0^1 \rho^5 d\rho \int_0^{\frac{\pi}{2}} \sin^3 \varphi d(\sin \varphi) \int_0^{\frac{\pi}{2}} \sin \theta d(\sin \theta) = \frac{1}{6} \cdot \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{48}. \end{aligned}$$

(3) 解: 由于积分区域 $\Omega$ 关于 $xOz$ 坐标平面对称, 因此 $\iiint_{\Omega} y dx dy dz = 0$ 且

$$\iiint_{\Omega} (y+z) dx dy dz = \iiint_{\Omega} z dx dy dz.$$

作广义球坐标变换:

$$\begin{cases} x = a\rho \sin \varphi \cos \theta \\ y = b\rho \sin \varphi \sin \theta \\ z = c\rho \cos \varphi \end{cases}$$

则 $\Omega$ 的边界曲面的方程为 $\rho=1$ , 直角坐标系下的空间区域 $\Omega$ 变为广义球坐标系下的区域:

$$\Omega_{\rho\varphi\theta} = \{(\rho, \varphi, \theta) \mid 0 \leq \rho \leq 1, 0 \leq \varphi \leq \frac{\pi}{2}, 0 \leq \theta \leq 2\pi\},$$

且

$$dx dy dz = \left| \frac{D(x, y, z)}{D(\rho, \varphi, \theta)} \right| d\rho d\varphi d\theta = abc \rho^2 \sin \varphi d\rho d\varphi d\theta.$$

故

$$\begin{aligned} \iiint_{\Omega} (y+z) dx dy dz &= \iiint_{\Omega} z dx dy dz = abc^2 \iiint_{\Omega_{\rho\varphi\theta}} \rho^3 \sin \varphi \cos \varphi d\rho d\varphi d\theta \\ &= abc^2 \int_0^1 \rho^3 d\rho \int_0^{\frac{\pi}{2}} \sin \varphi d(\sin \varphi) \int_0^{2\pi} d\theta \\ &= abc^2 \cdot \frac{1}{4} \cdot \frac{1}{2} \cdot 2\pi = \frac{\pi abc^2}{4}. \end{aligned}$$

(5) 解: 作球坐标变换:

$$\begin{cases} y = \rho \sin \varphi \cos \theta \\ z = \rho \sin \varphi \sin \theta \\ x = \rho \cos \varphi \end{cases}$$

则积分区域 $\Omega = \{(x, y, z) \mid y^2 + z^2 \leq x^2 \leq R^2 - y^2 - z^2, x \geq 0\}$ 在球坐标系下为

$$\Omega_{\rho\varphi\theta} = \{(\rho, \varphi, \theta) \mid 0 \leq \rho \leq R, 0 \leq \varphi \leq \frac{\pi}{4}, 0 \leq \theta \leq 2\pi\},$$

故

$$\begin{aligned}\iiint_{\Omega} (x^2 + y^2 + z^2) dx dy dz &= \iiint_{\Omega_{\rho\varphi\theta}} \rho^2 \cdot \rho^2 \sin \varphi d\rho d\varphi d\theta = \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{4}} \sin \varphi d\varphi \int_0^R \rho^4 d\rho \\ &= 2\pi \cdot \left(1 - \frac{\sqrt{2}}{2}\right) \cdot \frac{R^5}{5} = \frac{(2 - \sqrt{2})\pi R^5}{5}.\end{aligned}$$

7. (2) 解:  $V = \iiint_{\Omega} dx dy dz = \iint_D dx dy \int_{\sqrt{x^2+y^2}}^{6-x^2-y^2} dz = \frac{32\pi}{3}$ , 其中

$$\Omega = \{(x, y, z) | \sqrt{x^2 + y^2} \leq z \leq 6 - x^2 - y^2, (x, y) \in D\}, \quad D = \{(x, y) | x^2 + y^2 \leq 4\}.$$

(3) 解: 作球坐标变换  $\begin{cases} x = \rho \sin \varphi \cos \theta \\ y = \rho \sin \varphi \sin \theta \\ z = \rho \cos \varphi \end{cases}$  则所求立体  $\Omega$  在球坐标系中是长方体:

$$\Omega_{\rho\varphi\theta} = \{(\rho, \varphi, \theta) | a < \rho < b, 0 \leq \varphi \leq \frac{\pi}{4}, 0 \leq \theta \leq 2\pi\}.$$

$$\text{故 } V = \iiint_{\Omega} dx dy dz = \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{4}} d\varphi \int_a^b \rho^2 \sin \varphi d\rho = \frac{(2 - \sqrt{2})\pi}{3} (b^3 - a^3).$$

8. 解: (2) 作变换:

$$\begin{cases} u = xy \\ v = y \\ w = z \end{cases}$$

则新坐标系下的积分区域  $\Omega_{uvw}$  由  $w = u$  与  $u = v^2$ ,  $u = v$ ,  $w = 0$  围成, 表示为:

$$\Omega_{uvw} = \{(u, v, w) | 0 \leq v \leq 1, v^2 \leq u \leq v, 0 \leq w \leq u\}.$$

由于  $\left| \det \frac{\partial(u, v, w)}{\partial(x, y, z)} \right| = v$ , 因此原积分为

$$\begin{aligned}I &= \iiint_{\Omega_{uvw}} uvw^3 \cdot \frac{1}{v} du dv dw = \iiint_{\Omega_{uvw}} uw^3 du dv dw = \int_0^1 dv \int_{v^2}^v u du \int_0^u w^3 dw = \frac{1}{4} \int_0^1 dv \int_{v^2}^v u^5 du \\ &= \frac{1}{24} \int_0^1 dv (u^6) \Big|_{v^2}^v = \frac{1}{24} \left( \frac{1}{7} - \frac{1}{13} \right) = \frac{1}{364}.\end{aligned}$$

(3) 解: 作变换:

$$\begin{cases} u = x - y \\ v = x - z \\ w = x + y + z \end{cases}$$

直角坐标系下的空间区域  $\Omega$  变为  $O-uvw$  坐标系下的区域:

$$\Omega_{uvw} = \{(u, v, w) | 0 \leq u \leq 1, 0 \leq v \leq 1, 0 \leq w \leq 1\},$$

$$dx dy dz = \left| \det \frac{\partial(u, v, w)}{\partial(x, y, z)} \right|^{-1} du dv dw = \left| \det \begin{pmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 1 & 1 & 1 \end{pmatrix} \right|^{-1} du dv dw = \frac{1}{3} du dv dw,$$

故

$$\begin{aligned}
\iiint_{\Omega} (x+y+z) \cos((x+y+z)^2) dx dy dz &= \frac{1}{3} \iiint_{\Omega_{uvw}} w \cos w^2 du dv dw \\
&= \frac{1}{6} \int_0^1 du \int_0^1 dv \int_0^1 \cos w^2 dw^2 = \frac{\sin 1}{6}.
\end{aligned}$$